

Amirali Imanipannah

+98 910 776 2044 amir.imanipannah@gmail.com github.com/Amir-imani iocraft.ir Tehran, Iran

Professional Summary

Final-year Electrical Engineering student specializing in **Digital Electronics**, **FPGA-based Design**, and **Embedded Systems**. Proficient in high-speed **PCB Design** and autonomous robotics using **ROS2**. Experienced in implementing high-frequency hardware algorithms and integrating AI with classical control systems. Proven leadership through multiple Teaching Assistantships in microcontrollers and electronics.

Education

University of Isfahan Isfahan, Iran
Bachelor of Science in Electrical Engineering - Electronics Sep 2022 – July 2026 (Expected)

- **Overall GPA:** 16.50 / 20.00
- **Key Courses:** Microcontrollers, Digital Systems Design, Computer Architecture, Signals & Systems.

Research Interests

Digital Systems: FPGA-based design, Embedded Linux, Verilog/VHDL, and SoC.
Robotics: Autonomous Navigation, ROS2, and Real-time Systems.
AI in Hardware: Machine Learning on Microcontrollers (TinyML) and Computer Vision.

Technical Skills

Embedded & Robotics: STM32/STM8, RP2040, AVR, Raspberry Pi 5 (Embedded Linux), ROS2, ESP32, IOT Protocols.
Hardware Design: Altium Designer (Advanced), KiCad, Proteus, PSpice, SMD/DIP Soldering & Assembly.
Programming & Tools: C/C++, Verilog, Python, MATLAB, HTML/CSS/JS, Git, Linux CLI, SolidWorks.

Certifications

Professional PCB Design (Advanced level): Issued by IEEE Student Branch. 2024
Introductory PCB Design (Altium Designer): Issued by IEEE Student Branch. 2023

Teaching Experience

Teaching Assistant (TA) *University of Isfahan*

- **Microcontrollers (Spring 2026):** Guided 80+ students in AVR/STM32 interfacing and programming.
- **Electronics I (Spring 2026):** Conducted recitation sessions and assisted with transistor circuits.
- **Electric Circuits I (Fall 2025):** Conducted problem-solving workshops and network analysis tutorials.
- **Fundamentals of Programming (Fall 2025):** Taught C++ logic, structural debugging, and basic algorithms.

Selected Projects

Autonomous Mobile Robot with AI-Optimized Control *B.Sc. Thesis*

- Developed a ROS2/Raspberry Pi 5 robot integrating **SLAM**, **Nav2**, and edge AI for autonomous navigation.
- Implemented a **Neural Network-optimized PID controller** to solve dynamic stability, slope, and friction challenges.
- Designed custom power distribution and motor control PCBs utilizing Altium Designer.

High-Frequency CORDIC Processor Core *Digital Systems Design Project*

- Implemented a pipelined CORDIC algorithm on FPGA for hardware-level trigonometric operations, achieving a maximum operating frequency of **115.399 MHz**.

High-Performance STM32-Based Graphical HMI System *Independent Hardware Project*

- Designed an embedded system using an **STM32F407** to drive a 40-pin TFT LCD and capacitive touch panel.
- Developed a mathematical core for real-time graphing and matrix operations, suitable for industrial HMI use.

Analysis of MQ-Series Gas Sensors *Electronic Measurement Course*

- Investigated the electrochemical operation and internal electrode structures of MQ-family sensors.

Languages

English: Professional working proficiency (Preparing for IELTS). **Persian:** Native.